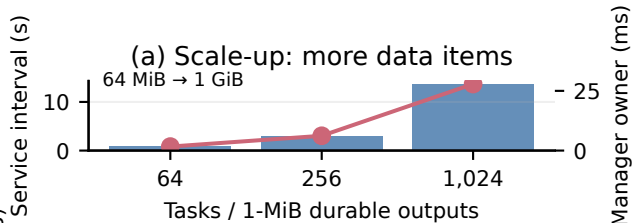


## Why scheduling and data management must not share one serialized service path

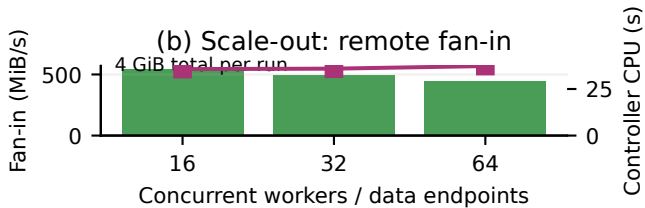
(a) Scale-up: more data items

64 MiB  $\rightarrow$  1 GiB



(b) Scale-out: remote fan-in

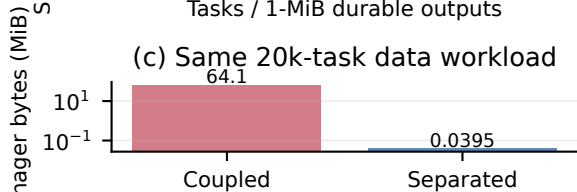
4 GiB total per run



(c) Same 20k-task data workload

64.1

0.0395



(d) Coupled path becomes the service bottleneck

160.4

10.0

